

Experiment - 14

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Course Code: MLA0305

Slot: B

Aim

To implement **Model-Based Reinforcement Learning** using data generation and a learned model to predict the next state of the environment.

Algorithm / Procedure

1. Create the CartPole environment.
2. Generate sample state-action transitions from the environment.
3. Store the generated data as training samples.
4. Train a regression model to learn the environment dynamics.
5. Use the trained model to predict the next state.
6. Compare the actual and predicted states.
7. Plot the comparison graph.

Code

```
import gymnasium as gym

# Test prediction
state, _ = env.reset()
action = env.action_space.sample()

prediction = model.predict([
    np.append(state, action)
])[0]

print("\nCurrent State:")
print(state)

print("\nPredicted Next State:")
print(prediction)
```

```
# Compare actual and predicted states
actual = []
predicted = []

state, _ = env.reset()

for _ in range(20):

    action = env.action_space.sample()

    next_state, reward, terminated, truncated, _ = env.step(action)

    pred = model.predict([
        np.append(state, action)
   ])[0]

    actual.append(next_state[0])
    predicted.append(pred[0])

    state = next_state

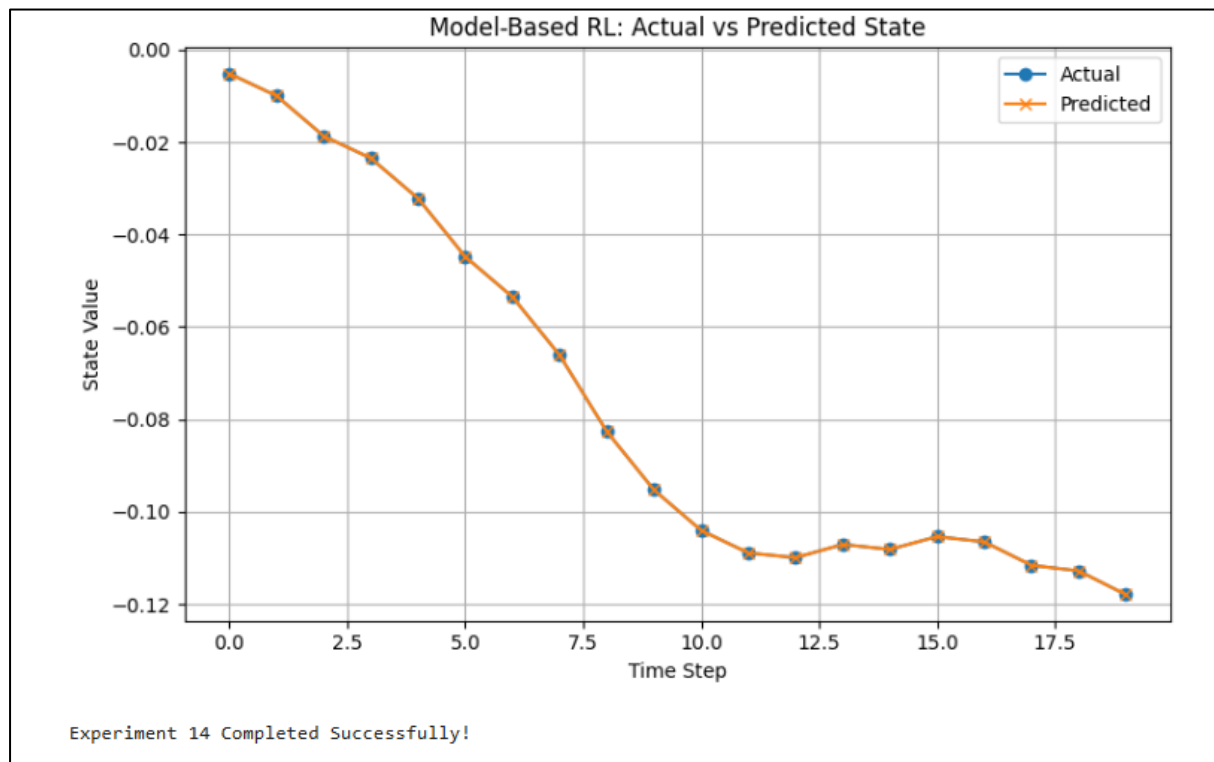
    if terminated or truncated:
        break

# Graph
plt.figure(figsize=(8, 5))

plt.plot(
    actual,
    marker="o",
    label="Actual"
)
```

```
plt.plot(  
    predicted,  
    marker="x",  
    label="Predicted"  
)  
  
plt.xlabel("Time Step")  
plt.ylabel("State Value")  
plt.title("Model-Based RL: Actual vs Predicted State")  
  
plt.legend()  
plt.grid(True)  
plt.tight_layout()  
plt.show()  
  
env.close()  
  
print("\nExperiment 14 Completed Successfully!")
```

Output



Result

A model-based reinforcement learning system was successfully implemented by generating environment data and learning the relationship between the current state, action, and next state. The predicted and actual states were compared graphically.